Board attributes and voluntary disclosure in an emerging economy: evidence from Nigeria

Robert W. Odewale

Tunku Puteri Intan Safinaz School of Accountancy,
College of Business,
Universiti Utara Malaysia,
Sintok, Kedah, Malaysia
Email: rowodewale@yahoo.com

Abstract: This study examines the effect of board attributes (board size, board composition and CEO duality) on the extent of voluntary disclosure using data for 237 firm-year observations from 75 firms listed on the Nigerian Stock Exchange from 2009 to 2012. The study constructs disclosure index score comprising 36 items. Using random-effects regression model, the result shows that CEO duality is negatively related to voluntary disclosure. This study also finds that board size and board composition do not have any significant relationship with voluntary disclosure. This study has implications for future researchers, regulators, and investors. Future researchers may find it interesting to examine board behaviour in order to understand the complexities of board operations as it affects their monitoring role. There is no evidence that the introduction of Corporate Governance Code by the Nigerian Securities and Exchange Commission has led to improvement in the voluntary disclosure made by listed companies. There is therefore the need for regulators to improve their enforcement and compliance mechanism at ensuring that listed companies comply with the disclosure requirements. It may also be appropriate that certain disclosures be made mandatory, since the management may not have the incentives to make such disclosures.

Keywords: board attributes; corporate governance; voluntary disclosure; emerging economy; board size; board composition; CEO duality; disclosure index; Nigeria.

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Biographical notes: Robert W. Odewale had lectured at the Polytechnic and University in Nigeria. He is a Chartered Accountant (Nigeria) and has a research doctorate. His main research interests are in the area of executive compensation, corporate governance, firm performance, and disclosure studies.

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1 Introduction

In modern complex corporations, there is information asymmetry between the executive management and firm shareholders as a result of separation of ownership and control, which results into agency conflicts (Healy and Palepu, 2001; Jensen and Meckling, 1976). John and Senbet (1998) assert that important decisions of the firm are made by the professional managers. Such decisions include the level of information to be made public to other firm stakeholders. Shareholders, potential investors and other relevant stakeholders use available information about firms to predict its long-term viability, thus enabling them to make wise investment decisions. The corporate scandals and financial crises of the recent past have lent credence to calls for improved corporate governance practice and increased managerial accountability (Adams et al., 2010; Correa and Lel, 2014; Hwang and Jung, 2016; Ntim et al., 2012a). This is because the managers possess superior information concerning the present and future operations of the firm more than all other stakeholders, since they are responsible for the day-to-day running of the firm's affairs. In fact, information asymmetry theory argues that managers are not inclined to disclose information to shareholders (Al-Janadi et al., 2016). Pierce (2011, p.4) points that "the level of transparency in all sections of the annual report is significantly lower in Nigeria than it is in many other countries, there is an opportunity to greater disclosure, greater transparency and there is a seeming reluctant to do that…". Further, Adegbite (2015, p.23) documents that one of the focus group respondents agrees that "we still need real board independence" in Nigeria.

This study is motivated by Gul and Leung (2004) who observe that corporate governance variables are less considered in voluntary disclosure studies. Second, few studies from Nigeria have focused on cross-sectional data. Third, the weak institutions in Nigeria provide an opportunity to examine the effectiveness of board variables in monitoring managerial actions. Four, studies on voluntary disclosure on Nigeria are few while none has considered how boards influence the level of such disclosure. There are several studies on boards and voluntary disclosure from developed economies but their findings cannot be generalised for an emerging economy like Nigeria because of the weak institutions and low investor protection rights (Okike, 2007; ROSC, 2008, 2011). It therefore becomes imperative to examine how boards influence the level of voluntary disclosure in Nigeria since the country is one of the fastest growing economies in the world.

To control for this information asymmetry between the shareholders (principal) and the managers (agent) that is addressed as an agency problem, there is need for more disclosure of firm information (Verrecchia, 2001). The financial fraud that characterised corporate giants like Enron, WorldCom, and Global Crossing has increased the attention on corporate voluntary disclosure and financial reporting. Consequently, stakeholders now demand for corporate transparency and accountability from the executive management in the form of increased disclosure of information in the annual report (Ntim et al., 2012a, 2012b).

The increased disclosure requirement is an effort to reduce information asymmetry between managers and stakeholders. Disclosure of information by firms has its attendant benefits. For example, Healy and Palepu (2001) argue that firm's management uses financial reporting and disclosure as the medium for communicating to outside investors the performance and governance of the firm. Further, it is argued that firms make voluntary disclosure in an effort to reduce information risk that could negatively impact

the firm's stock prices (Graham et al., 2005; Healy and Palepu, 2001). This is because executives believe that ambiguous information or a reputation for not consistently making precise and accurate information available to the market could lead to under-pricing the firm's stock (Graham et al., 2005).

The board is one of the firm's key internal corporate governance mechanisms that has been discussed extensively in the corporate governance literature (Fama and Jensen, 1983; Guest, 2008; Şener et al., 2011). It is therefore of interest to examine the effects of board attributes on voluntary disclosure. Board attributes such as board size, board composition and CEO duality have been shown to influence organisational outcomes (Allegrini and Greco, 2013; Munisi et al., 2014; Samaha et al., 2012). Therefore, understanding how these board attributes shape firms' disposition to voluntary disclosure in an emerging economy will have implication for academic researchers, policy makers, and investors, especially in Nigeria.

The purpose for this study is threefold. First, it examines whether board size is associated with voluntary disclosure. Second, it examines whether board composition is associated with increased voluntary disclosure. Third, it examines whether CEO duality is associated with decrease in voluntary disclosure. Past studies on voluntary disclosure from Nigeria include Adelopo (2011) and Nyahas et al. (2018), this paper differs from prior ones in terms of:

a the sample period 2009 to 2012

b use of board attribute variables

c use of panel dataset.

Adelopo (2011) excluded corporate governance variables from his study while Nyahas et al. (2018) used cross-sectional data. This study complements this line of research by considering how board size, board composition and CEO duality influence voluntary disclosure. Second, by using panel dataset this study captures changes in voluntary disclosure over time. Available information suggest that this is the first study to examine the influence of board attributes on the level of voluntary disclosure in Nigerian listed companies (NLCs) using panel dataset.

It is anticipated that undertaking this study in a country like Nigeria may uncover significant policy implications to other emerging markets with weak institutions. The study also contributes to the literature by applying the results of empirical analysis with insight from agency theory and information asymmetry theory. This study makes additional contribution to the literature on the influence of board attributes on voluntary disclosure in an emerging economy with weak institutional framework. Using a sample of 75 firms listed on the Nigerian Stock Exchange (NSE) with 237 firm-year observations during the period 2009 to 2012, the study shows that CEO duality is associated with lower level of voluntary disclosure by firms. An interesting finding is that board composition does not significantly influence corporate voluntary disclosure. This could be an indication that the boards are unable to determine what is considered appropriate behaviour for the management (Tosi et al., 2003). There is evidence that board size has a benign effect on voluntary disclosure in NLCs.

The remainder of this paper is structured as follows: first, section two provides the review of relevant literature and development of hypotheses. Second, section three provides a description of the research design and methodology used for the study. Third,

section four presents the empirical results. Finally, section five contains the concluding comments.

2 Literature review and hypotheses development

The corporate scandals and failures of the first decade of this century in the USA, Europe, Australia and other countries, that were blamed on poor corporate governance practice, have made legislative and regulatory authorities from several countries turn their focus on the board of directors in an effort to improve corporate governance practice (Boone et al., 2007; Denis and McConnell, 2003). Hwang and Jung (2016) show that good corporate governance is associated with improved firm performance. The Nigerian Government through the Securities and Exchange Commission, Nigeria (SECN) and the Corporate Affairs Commission (CAC) initiated efforts at improving the country's corporate governance practice by publishing the Corporate Governance Code for public companies in Nigeria (CG Code) in 2003 with a revised code in 2011. Part of the efforts include the recommendation that the board should consist of more non-executive directors, separation of the position of the chief executive officer (CEO) and that of the chairman, and a minimum board size of five directors. These efforts were geared at ensuring that the board is effective in the discharge of its monitoring duty and responsibility for financial reporting.

From the international perspective, past studies conducted to investigate the relationship between board attributes and voluntary disclosure have provided conflicting results (Akhtaruddin et al., 2009; Allegrini and Greco, 2013; Arcay and Vázquez, 2005; Barako et al., 2006; Chen and Jaggi, 2000; Cheng and Courtenay, 2006; Eng and Mak, 2003; Lim et al., 2007; Ghazali and Weetman, 2006; Samaha et al., 2012, 2015).

2.1 Board of directors

In market economies, there is separation of ownership and control in firms where the shareholders have delegated the control of the firm to professional managers. If the professional managers are left to themselves, they may not act in the best interest of the shareholders as they may likely exhibit opportunistic behaviour to the detriment of the shareholders. The diffused nature of shareholdings in public firms makes it practically impossible for all shareholders to be involved in monitoring the activities of the professional managers, since large groups have been found to have problem of ease of coordination, consensus in reaching decisions and free riding (Cahan et al., 2005; John and Senbet, 1998; Laksmana, 2008; Linck et al., 2008; Ntim et al., 2015). Monitoring of the managers is assigned to another agent of the shareholders addressed as the board of directors. Section 244 of the Companies and Allied Matters Act (CAMA) 1990 as amended provides the legal backing for the creation of board of directors of firms registered in Nigeria.

The importance of the existence of the board in corporate organisations is not in doubt as it is widely acclaimed it has a role to play in mediating between the diffused shareholders and the professional managers who have taken control of the day-to-day running of the firm (Fama and Jensen, 1983; Jensen and Meckling, 1976). The board has been described as an internal corporate governance mechanism of the firm that is put in place to serve as the representatives of the diffused shareholders to prevent the executives

from exhibiting opportunistic behaviour (Baysinger and Hoskisson, 1990; Boyd, 1994; Brick et al., 2006; Dehaene et al., 2001; Guest, 2008; Hermalin and Weisbach, 2003; Hillman et al., 2000; Jensen, 1993; Lefort and Urzúa, 2008; Ozdemir and Upneja, 2012). The board performs various functions as policy and strategy formulation, hiring and firing executives, determination of executive compensation, advising, and monitoring the activities of management (Boyd, 1994; Brick et al., 2006; Jensen, 1993; Linck et al., 2008; Zahra and Pearce, 1989). They also have the duty to reduce information asymmetry between the shareholders and firm managers in the form of enhancing financial reporting through adequate disclosure of information to the shareholders (SECN, 2011).

Hillman et al. (2000) asserts that board structure (composition and size) is a function of uncertainty and environmental dependency facing the firm. Therefore, there cannot be a uniform board structure for all firms. Consistent with Hillman et al. (2000), Coles et al. (2008) report that the firm's complexity determines the board structure, as complex firms are found to have large boards and majority outside directors. They associated large boards to the advisory needs of the firm resulting from its complexity. Zahra and Pearce (1989) identified control, service and strategy as the main functions of boards. The control function is exercised over the activities of management by monitoring, hiring, firing, and determining the compensation of the executive (Masulis et al., 2012; Şener et al., 2011). On the service role of the board, non-executive directors bring in their expertise to bear on the operations of the firm, since they have different backgrounds, by providing useful advice to the executive. Since the creation of the board, the often-asked question has been whether the board has been effective or otherwise in the performance of its monitoring role.

Jensen (1993) asserts that the board of directors remain ineffective to monitor the CEO because of the board culture that does not allow for antagonism. Jensen views this culture as one that needs to be jettisoned if the board must perform its assigned duty effectively. The board is endogenously determined (Hermalin and Weisbach, 1998) by recommendations from incumbent board members and primarily by the CEO through the nominating committee. It is therefore not unlikely that the new board members will belong to the directors' social network. If this situation holds, such board may not be able to perform its monitoring role effectively, as they would like to avoid confrontation as much as possible. The outcome will be the making of a CEO that has strong influence over the board that may eventually lead to his entrenchment. Kirchmaier and Stathopoulos (2008) demonstrate that the CEOs that have access to large social networks will manifest opportunistic behaviour to the detriment of the firm. They find that CEOs' social network impact on firm performance is negative, as highly connected CEOs are able to evade proper board monitoring.

According to Fama and Jensen (1983), more outside directors on boards do the monitoring of the management in situations where the takeover market is weak. Yermack (1996) shows that board size has implications for corporate governance, and Core et al. (1999) provide evidence that CEO duality is a proof of weak corporate governance system. In summary, it is the responsibility of the board to ensure that the code of corporate governance is adopted and implemented by the management. If this function is dutifully performed it is expected to produce increased voluntary disclosure and constrain the management from exhibiting opportunistic behaviour. Consistent with the research objectives, the following discussions lead to the statement of the study's hypotheses.

2.1.1 Board size

It is argued that board size is a determinant of board of directors' effectiveness (Allegrini and Greco, 2013; Jensen, 1993) and resource dependence theory predicts that directors with expertise will form a resource pool for the firm (Chen and Roberts, 2010; Zahra and Pearce, 1989). Prior research suggests that a firm's board size will depend on the size and complexity of its activities (Linck et al., 2008). However, there are competing views as regards the board size that is appropriate to maintain quality board that ensures adequate monitoring of managerial actions. Some studies argue that large boards are well positioned to do proper monitoring of the management (Anderson et al., 2004) and satisfy the advisory needs of complex firms (Coles et al., 2008).

In contrast, other scholars argue that large boards will be less effective in monitoring management activities. For example, Yermack (1996) provides evidence that large boards are not quick in making decisions. In addition, large boards are prone to enter into compromises to arrive at decisions just as cohesion is also found to be low, as they are also faced with difficulty of coordination (Cahan et al., 2005; Cheng, 2008). Such decisions within the purview of the board include extent of information disclosure in the annual reports.

In voluntary disclosure studies, board size is shown to be positively related to voluntary disclosure (Akhtaruddin et al., 2009; Al-Janadi et al., 2013; Allegrini and Greco, 2013; Laksmana, 2008; Ntim et al., 2012b; Samaha et al., 2015). In another strand of literature, board size is reported to show insignificant relationship with voluntary disclosure (Arcay and Vázquez, 2005; Cheng and Courtenay, 2006; Samaha et al., 2012). Following the mixed findings from prior studies, this study provides the following non-directional hypothesis.

H1 There is a significant relationship between board size and voluntary disclosure.

2.1.2 Board composition

The board is made up of both the executive and non-executive directors who Şener et al. (2011) describe as boundary spanners that aligns the interest of the shareholders with those of the managers. Establishment of independent boards has become the norm in the corporate governance codes of several countries that adopt the Anglo-American corporate governance model with single-tier board system. This is because the presence of non-executive directors in firms is usually seen as a means of encouraging transparency from the executive management (Hillman et al., 2000; Ozdemir and Upneja, 2012). Eng and Mak (2003) argue that outside directors who are pressure resistant may have incentives to require firms to make increased disclosure of information to investors. Under the agency theory, Fama and Jensen (1983) argue that independent boards will curb managerial opportunism as they will become effective monitors of management activities.

Prior studies on voluntary disclosure provide evidence of independent boards' representation of shareholders' interests which is consistent with the argument of Fama and Jensen (1983). For example, Arcay and Vázquez (2005) show that board independence improves the monitoring ability of the board and enhances the quality of information made available to shareholders. Akhtaruddin et al. (2009) also document a positive association between the proportion of independent directors and extent of voluntary disclosure while Samaha et al. (2012) show that the presence of outside

directors is associated with overall corporate governance disclosure. This is consistent with existing evidence as documented by Chen and Jaggi (2000) and Samaha et al. (2015) of a positive effect of proportion of independent directors on mandatory disclosures and voluntary disclosure, respectively.

In other settings, outside directors have been shown to demonstrate interest in protecting the interests of the shareholders. For example, Bhagat and Bolton (2008) find that board independence is a signal for good corporate governance practice in a firm, as they report a positive relationship between board independence and management turnover when a firm reports poor performance. Similarly, Lefort and Urzúa (2008) find that firms derive greater value by increasing the proportion of outside directors on their boards. They also report that in an attempt to improve their corporate governance rating, firms that harbour the agency conflict situation will admit professional directors to their boards.

An important section of the 2011 code is the requirement that the board should comprise of more non-executive directors. It is therefore expected that compliance with this recommendation may increase board effectiveness and enhance the quality of corporate governance in firms. This study therefore argues that improvement in corporate governance quality is expected to enhance voluntary disclosure in firms. Consistent with the prediction of the agency theory and prior literature discussed above, this study expects that board independence may lead to effective monitoring of management to improve the extent of voluntary information disclosure by firms. The following hypothesis is therefore developed to test the relationship in this study.

H2 There is a positive relationship between board composition and voluntary disclosure.

2.1.3 CEO duality

The agency theory posits that when the position of both the CEO and chairman is occupied by an individual it confers immense power on such person that enables him to evade effective board monitoring and render independent board ineffective (Fama and Jensen, 1983; Gul and Leung, 2004). Laksmana (2008) argues that the CEO could influence the extent of information disclosure made by firms. It is also argued that CEO duality inhibits transparency and leads to high information asymmetry (Allegrini and Greco, 2013), which invariably leads to lower level of information disclosure (Gul and Leung, 2004).

Past disclosure studies have shown that CEO duality is associated with lower voluntary disclosure. For example, Gul and Leung (2004) using a 1996 cross-sectional sample of 385 Hong Kong companies provide evidence of a negative relationship between CEO duality and voluntary disclosure. Consistent with Gul and Leung (2004), Allegrini and Greco (2013) document a negative association between the variables with evidence from Italy. In the African context, Samaha et al. (2012) show that power concentration in an individual leads to lower level of voluntary disclosure with evidence from Egypt. In addition, using a sample of 64 empirical studies in a meta-analytic study, Samaha et al. (2015) document a significant negative effect of CEO duality on voluntary disclosure.

In line with global practice of good corporate governance, this issue was aptly captured by the 2011 code, which recommends that different individuals occupy both positions so as to avoid power concentration in an individual. Consistent with prior studies and corporate governance good practice, it is therefore expected that the

combination of the CEO position with that of the chairman will lower voluntary information disclosure. Consequently, in consonance with agency theory, the following hypothesis is tested.

H3 There is a negative relationship between CEO duality and voluntary disclosure.

3 Research design and methodology

3.1 Sample and data description

This study used unbalanced panel dataset from firms listed on the NSE during the period 2009 to 2012. This period coincides with a year after the SECN set up a new committee to revise the 2003 CG Code and a year after the publication of the 2011 CG Code. For the purpose of estimation unbalanced panel dataset does not pose any problem (Conyon and He, 2011). The data for this study were accessed from different sources. First, the relevant corporate governance data were obtained from the annual reports accessed through firm websites, africanfinancials.com, and hard copies. Second, nigeriagalleria.com and NSE Factbook 2011 provided information on the firms' age since listing on the NSE.

The data for all relevant information were hand-collected since there is no database available from where to access such information, unlike what obtains in the developed economies. This is one of the limiting factors for meaningful research in developing economies especially from Sub-Saharan Africa. The study sample consists of 119 firms listed on the NSE at the beginning of the selection process. The banks, insurance firms and other financial institutions were excluded because of their peculiar reporting nature. 88 firms were left after the initial screening process; this was further reduced due to unavailability of annual report for the study period. The final sample consists of 75 firms with 239 firm-year observations and was further reduced to 237 firm-year observations after deleting the outliers.

3.2 Variables and regression model

3.2.1 Dependent variable

The dependent variable of study is the voluntary disclosure index (VDI). Voluntary disclosure score is a dichotomous variable that takes the value of 1 if an item is disclosed or 0 otherwise. For each firm-year observation the VDI is the aggregate score obtained by a firm divided by the maximum score obtainable. In this study, disclosure index is used to measure the level of voluntary disclosure based on the information accessed from the annual reports similar to previous studies (Al-Bassam et al., 2018; Allegrini and Greco, 2013; Cheng and Courtenay, 2006; Eng and Mak, 2003, Gul and Leung, 2004). The disclosure items used by Meek et al. (1995) and Leventis and Weetman (2004) were adjusted for constructing the disclosure index to suit Nigeria's business environment as legislation, culture and politics have been shown to affect corporate governance and disclosure behaviour of firms.

Content analysis is used in constructing the disclosure index as this has been used primarily in the disclosure index studies (Beattie et al., 2004). Content analysis is a "research technique for making replicable and valid inferences from texts (or other

meaningful matter) to the contexts of their use" [Krippendorff, (2004), p.18]. Following Smith and Taffler (2000), this study adopts both the word-oriented and meaning-oriented approaches as suggested by Weber (1990) and Krippendorff (2004) for scoring the disclosure index. For the word-oriented analysis, the occurrence of the word or phrase connoting the intended meaning is regarded as sufficient disclosure while the meaning-oriented approach considers the sentence(s) before making inference concerning the disclosure item.

Overall, disclosure index score sheet was constructed comprising 36 items as shown in Appendix 1. Prior researchers have used weighted disclosure indices (Barako et al., 2006; Botosan, 1997) while others chose the unweighted disclosure indices for their studies (Adelopo, 2011; Aljifri, 2008; Gul and Leung, 2004; Hossain and Hammami, 2009; Kamel and Shahwan, 2014). Using weighted and unweighted disclosure index methods, Barako et al. (2006) arrived at the same set of results and conclusions. Since users of financial information have different needs, their assessment of different disclosure elements will differ and as such, this study opts for the unweighted approach to maintain objectivity.

3.2.2 Independent variables

Board size (BDSIZE), board composition (BDCOMP), and CEO duality (CEOCH) are considered as independent variables of interest in this study. Board size is represented as the number of both the executive and non-executive directors on board (Al-Bassam et al., 2018; Allegrini and Greco, 2013). BDCOMP is measured as the percentage of non-executive directors on board (Akhtaruddin et al., 2009; Barako et al., 2006; Eng and Mak, 2003). CEOCH is a dichotomous variable that is assigned the value of '1' if both positions of the CEO and chairman are occupied by an individual, otherwise '0' (Allegrini and Greco, 2013; Haniffa and Cooke, 2002; Samaha et al., 2012).

3.2.3 Control variables

The review of previous studies on voluntary disclosure informed the choice of control variables included in this study. These control variables are CEO shareholding (CEOSH), directors' shareholding (DIRSH), block shareholders (BLKSH), foreign shareholders (FORSH), auditor type (AUDT), listing age (LSTAG), firm performance (PERF), firm size (SIZE), leverage (LEV), and sector (Adelopo, 2011; Akhtaruddin et al., 2009; Allegrini and Greco, 2013; Collett and Hrasky, 2005; Hossain and Hammami, 2009; Leventis and Weetman, 2004; Lim et al., 2007; Samaha et al., 2012).

CEO shareholding is the percentage of shares held by the CEO to the total number of shares outstanding (Linck et al., 2008). Directors' shareholding represents the proportion of shares held by the directors to the total number of shares outstanding (Ghazali and Weetman, 2006). Voluntary disclosure studies have documented incongruous results on the relationship between voluntary disclosure and director's shareholding (Eng and Mak, 2003; Ghazali and Weetman, 2006; Samaha et al., 2012). The measure for block shareholders is the total shareholding by shareholders that hold a minimum of 5% of total firm shares outstanding (Al-Bassam et al., 2018; Eng and Mak, 2003; Odewale and Kamardin, 2015). CAMA, 1990 regards 5% shareholding as substantial in NLCs. Foreign shareholders represent the percentage of shares held by foreign investors in the firm (Barako et al., 2006).

Auditor type is a dummy variable that is assigned the value of '1' if the firm is audited by any of the Big 4 audit firms and '0' otherwise (Akhtaruddin and Haron, 2010; Barako et al., 2006). The Big 4 audit firms are among the reputable audit firms acknowledged internationally. Their certification of annual reports indicates a high-quality audit that is generally expected to be associated with the level of information disclosure by the audited firms (Adelopo, 2011; Akhtaruddin and Haron, 2010). There is the contention that the age of the firm will be associated with the extent of disclosures made by firms as older firms may likely make more disclosures than younger ones (Hossain and Hammami, 2009). Hossain and Hammami (2009) used the age of the firm since formation, but this study adopts the listing age similar to that of Haniffa and Cooke (2002). Prior research provides evidence of a positive relationship between firm performance and voluntary disclosure (Akhtaruddin and Haron, 2010; Lim et al., 2007) suggesting that profitable firms have the incentives to make higher disclosure in the annual reports than the loss-making firms (Meek et al., 1995). Consistent with previous voluntary disclosure studies, return on assets (ROA) is used to measure firm performance (Adelopo, 2011; Allegrini and Greco, 2013).

Log of total assets is used in this study to measure firm size in line with previous voluntary disclosure studies (Adelopo, 2011; Allegrini and Greco, 2013; Ho et al., 2013). Large firms are expected to make higher information disclosure in the annual reports than their smaller counterparts as they will be able to bear costs associated with such disclosures (Adelopo, 2011; Akhtaruddin and Haron, 2010). Leverage is total liabilities deflated by total assets (Al-Bassam et al., 2018; Allegrini and Greco, 2013; Eng and Mak, 2003). It is argued that high levered firms may be inclined to make higher disclosures to satisfy their creditors' need and minimise legal risk (Aljifri, 2008; Lim et al., 2007) as debt holders have the incentive to monitor and discipline the managers (Adelopo, 2011). Past studies that have included leverage in voluntary disclosure studies report mixed findings (Adelopo, 2011; Aljifri, 2008; Akhtaruddin and Haron, 2010; Barako et al., 2006; Gul and Leung, 2004). The sector measure is a dichotomous variable that takes the value of '1' if the firm is in that sector or '0' otherwise. The four sectors used for this study as shown in Table 2 consist of consumer goods and services, industrial goods and construction, oil and chemicals, and others (Barako et al., 2006; Leventis and Weetman, 2004; Meek et al., 1995; Gul and Leung, 2004).

3.3 Empirical model

This study employed the fixed-effects panel data methods to control for unobserved firm characteristics (Wooldridge, 2002). The F-test confirmed the existence of firm-fixed effects at 5% level. The Breusch Pagan Lagrange Multiplier test confirmed the suitability of the random-effects model at 5% level. These tests confirmed the preference for both fixed effects and random effects over pooled OLS regression. Hausman specification test was conducted to determine whether fixed effects or random effects are more appropriate. Hausman test failed to reject the hypothesis that the random effects estimates are inconsistent, suggesting the appropriateness of the random effects specification. The following regression model is estimated to examine the relationship between board attributes and voluntary disclosure.

$$VDI_{it} = \beta_0 + \beta_1 BDSIZE_{it} + \beta_2 BDCOMP_{it} + \beta_3 CEOCH_{it} + \beta_4 CEOSH + \beta_5 DIRSH_{it} + \beta_6 BLKSH_{it} + \beta_7 FORSH_{it} + \beta_8 AUDT_{it} + \beta_9 LSTAG_{it} + \beta_{10} PERF_{it} + \beta_{11} SIZE_{it} + \beta_{12} LEV_{it} + \beta_{13} CONGS_{it} + \beta_{14} INDGC_{it} + \beta_{15} OILC_{it} + \beta_{16} OTHS_{it} + \varepsilon_{it}$$

where

VDI	voluntary disclosure index
BDSIZE	board size
BDCOMP	board composition
CEOCH	CEO duality
CEOSH	CEO shareholding
DIRSH	directors share holding
BLKSH	block shareholders
FORSH	foreign shareholders
AUDT	auditor type
LSTAG	listing age
PERF	firm performance
SIZE	firm size
LEV	leverage
CONGS	consumer goods and services sector
INDGC	industrial goods and construction sector
OILC	oil and chemicals sector
OTHS	others sector
ε	random error term
i	firm 1 to n
t	year 1 to t.

4 Empirical results

4.1 Descriptive statistics

Descriptive statistics for the dependent variable is shown in Table 1 while the descriptive statistics for all variables that constitute the full sample is presented in Table 2. The mean VDS increased from 47.80% in 2009 to 52.20% by 2010, but show a steady decline to 49.20% by 2012. Overall, the result indicates an average voluntary disclosure of 49.70% having a minimum and maximum disclosure of 30.60% and 75% respectively. This is an

improvement over the mean (44%) for NLCs reported by Adelopo (2011). The average board size is 8.6 with a minimum of 5 and maximum of 15. On average, 71.10% of the board members are outside directors. The ratio of outside directors to total directors has the highest percentage of 93.33% with the lowest at 20%. Mean CEO duality is 3.40% indicating that 96.60% of sample observations have separate individuals for the post of the CEO and chairman.

Table 1 Descriptive statistics of voluntary disclosure score for the period 2009 to 2012

Year	*No.*	*Mean*	*Minimum*	*Maximum*	*Std. dev.*
2009	58	0.478	0.306	0.722	0.097
2010	60	0.522	0.361	0.722	0.089
2011	61	0.497	0.333	0.750	0.106
2012	58	0.492	0.333	0.722	0.112
All	237	0.497	0.306	0.750	0.102

Table 2 Descriptive statistics for study variables (N = 237)

Variables	*Mean*	*Minimum*	*Maximum*	*Std. dev.*
VDS	0.497	0.306	0.750	0.102
BDSIZE	8.595	5	15	2.132
BDCOMP	0.711	0.2	0.933	0.149
CEOCH	0.034	0	1	0.181
CEOSH	0.037	0	0.4	0.093
DIRSH	0.173	0	0.914	0.228
BLKSH	0.301	0	0.964	0.273
FORSH	0.237	0	0.813	0.290
AUDT	0.7597	0	1	0.428
LSTAG	23.080	0	47	12.623
PERF	0.108	–9.438	0.578	0.126
SIZE	16.432	12.032	20.328	1.623
LEV	0.592	0.116	1.521	0.228
SECTORS:				
CONGS	0.565	0	1	0.497
INDGS	0.177	0	1	0.383
OILC	0.143	0	1	0.351
OTHS	0.127	0	1	0.370

4.2 Correlation analysis

The correlation matrix is shown in Table 3. The result shows that VDI is positively correlated with the BDSIZE (r = 0.250) but does not show any significant correlation with BDCOMP and CEOCH, where r is the correlation coefficient. On the control variables, FORSH (r = 0.178), AUDT (r = 0.196), PERF (r = 0.364), and SIZE (r = 0.410) show significant positive correlation with VDS while CEOSH (r = –0.155),

DIRSH (r = –0.262), BLKSH (r = –0.260) and OTHS sector (r = –0.174) show significant negative correlation with VDS. The study does not find significant correlation between VDS and LSTAG, LEV, and sectors (CONGS, INDGC and OIL).

For the independent variables, there is no significant correlation between BDSIZE and BDCOMP. However, CEOCH is significantly negatively correlated with both BDSIZE and BDCOMP. The result further indicates that firms that report inverse association between BDCOMP and DIRSH are more likely to report lower voluntary disclosure. This is also true for firms where there is inverse association between BDSIZE and DIRSH. A firm with negative association between BDSIZE and CEOSH will more likely experience lower voluntary disclosure. The correlation matrix shows the highest correlation coefficient (r = –0.639) exists between BLKSH and FORSH.

4.3 Multivariate analysis

4.3.1 Main regression analysis and discussion

The random effects regression result presented in Table 4 shows the relationship between voluntary disclosure and the predictor variables (independent and control variables). The estimate of BDSIZE on the voluntary disclosure is negative and statistically insignificant. The results fail to support H1 that there is a significant relationship between board size and voluntary disclosure. The implication of this result is that the board size is not a determinant of the level of voluntary disclosure. This is consistent with the view that there is no one-size-fits-all for firms (Guest, 2008; Sanda et al., 2011), and that board size should depend on the complexity of the firms' operations (Linck et al., 2008). The result is similar to the findings by previous researchers from both developed and emerging economies that document non-association between the variables (Cheng and Courtenay, 2006; Samaha et al., 2012). For example, Cheng and Courtenay (2006) show that there is no association between board size and voluntary disclosure from their study sample of firms listed on Singapore Stock Exchange. Similarly, with evidence from Egypt, Samaha et al. (2012) document that board size is not a determinant of level of voluntary disclosure. The explanation for this finding could be that the board members are pressure sensitive. It could also be that the directors do not want to be seen as being antagonistic, since there is a board culture that does not allow for conflict (Jensen, 1993).

The estimate of BDCOMP on the voluntary disclosure is positive and statistically insignificant. The H2 that there is a positive relationship between board composition and voluntary disclosure is not supported. The implication of this finding is that outside directors have benign effect on voluntary disclosure and they are not better monitors than inside directors. This insignificant relationship could be an indication that the outside directors are indifferent to disclosure matters in the annual reports as they have access to their needed information from the management. Another explanation for this insignificant finding could be as a result of non-separation of outside directors into whether they are gray[1] or independent directors in this study. Furthermore, the significant compliance with the CG Codes that boards should comprise more non-executive directors (mean = 71.10%) could be the reason for this finding. The result is consistent with prior studies (Haniffa and Cooke, 2002; Ghazali and Weetman, 2006) but contrasts the report by Barako et al. (2006) and Samaha et al. (2012).

Table 3 Pearson correlation matrix (N = 237)

		1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*
VDI	1	1.000																
BDSIZE	2	0.250***	1.000															
BDCOMP	3	–0.059	0.043	1.000														
CEOCH	4	–0.086	–0.283***	–0.218***	1.000													
CEOSH	5	–0.155**	–0.249***	–0.185***	0.213***	1.000												
DIRSH	6	–0.262***	–0.302***	–0.165***	0.011	0.542***	1.000											
BLKSH	7	–0.260***	–0.269***	0.181***	–0.085	0.194***	0.408***	1.000										
FORSH	8	0.178***	0.186***	–0.131**	0.041	–0.186***	–0.284***	–0.639***	1.000									
AUDT	9	0.196***	–0.019	–0.118*	0.105	–0.209***	–0.233***	–0.163**	0.206***	1.000								
LSTAG	10	0.018	–0.204***	–0.079	0.162**	–0.041	–0.070	–0.300***	0.216***	0.110*	1.000							
PERF	11	0.364***	0.153**	–0.062	0.019	–0.126*	–0.268***	–0.129**	0.165**	0.143**	0.066	1.000						
SIZE	12	0.410***	0.487***	–0.099	–0.046	–0.233***	–0.324***	–0.247***	0.318***	0.269***	–0.150**	0.247***	1.000					
LEV	13	–0.080	–0.077	0.006	0.216***	–0.122*	0.045	0.026	0.019	0.075	0.172***	–0.333***	–0.039	1.000				
CONGS	14	0.050	0.077	–0.001	–0.025	0.109*	–0.122*	–0.101	0.008	–0.195***	–0.013	0.126*	–0.097	–0.208***	1.000			
INDGS	15	0.047	–0.047	0.078	–0.087	0.093	0.102	0.185***	–0.116*	0.080	0.037	–0.054	0.046	–0.058	–0.529***	1.000		
OILC	16	0.029	0.016	–0.219***	0.190***	–0.160**	–0.178***	0.162**	0.132**	0.230***	0.074	0.122*	0.203***	0.255***	–0.467***	–0.190***	1.000	
OTHS	17	–0.174***	–0.096	0.131**	–0.064	–0.195	0.291***	0.060	0.028	–0.021	–0.061	–0.251***	–0.081	0.108*	–0.392***	–0.159**	–0.043	1.000

Note: *$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

Consistent with H3, the result supports the prediction that there is a negative relationship between CEO duality and voluntary disclosure at 5% level ($\beta = -0.063$, $p = 0.022$). This finding of lower voluntary disclosure when the CEO is also the board chair is similar to the finding of Gul and Leung (2004) and Samaha et al. (2015). The result however contrasts the finding of Cheng and Courtenay (2006) who document an insignificant relationship between the variables. The implication of this result is that CEO duality weakens boards' monitoring role, inhibits transparency and leads to high information asymmetry between the CEO and the shareholders. This finding provides support for the recent efforts by SECN at ensuring that separate individuals occupy the position of the CEO and chairman. This is consistent with the view that concentration of power in an individual in firms could harm the shareholders (Gul and Leung, 2004) and the individual may be able to evade effective board monitoring (Fama and Jensen, 1983). It is also consistent with the argument that separate individuals should occupy the post of the CEO and that of the chairman (Jensen, 1993).

Table 4 Random-effects regression (with robust standard errors) results for voluntary disclosure index

Variable	*Expected sign*	*Estimates*	*z-statistic*	*p-value*
Constant		0.119	1.08	0.282
BDSIZE	+/–	–0.003	–0.83	0.406
BDCOMP	+	0.015	0.35	0.724
CEOCH	–	–0.063**	–2.28	0.022
CEOSH		0.007	0.08	0.939
DIRSH		–0.001	–0.02	0.985
BLKSH		–0.080**	–2.02	0.044
FORSH		–0.027	–0.78	0.433
AUDT		0.026	1.36	0.175
LSTAG		0.000	0.10	0.917
PERF		0.179***	2.62	0.009
SIZE		0.023***	3.70	0.000
LEV		0.029	1.00	0.318
SECTORS:				
CONGS		0.003	0.19	0.851
INDGC		–0.002	–0.06	0.949
OILC		–0.030	–1.18	0.238
OTHS		–0.015**	–2.35	0.019
F-statistics		215.23***		
Adjusted R^2		n.a		
Number of observations		237		
Wald test of joint significance of board structure variables			F-statistics	5.57
			p-value	0.1346

Note: *$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

For the control variables, the proportion of shareholding by block shareholders ($\beta = -0.080$, $p = 0.044$) is related to lower voluntary disclosure. Voluntary disclosure also decreases when a firm belongs to others sector ($\beta = -0.015$, $p = 0.019$). On the other hand, the result shows that high performing firms have the incentive to make higher voluntary disclosure ($\beta = 0.179$, $p = 0.009$). Similarly, the result indicates large firms make increased disclosures in their annual report ($\beta = 0.023$, $p = 0.000$). The analysis shows that the other control variables; CEOSH, DIRSH, FORSH, AUDT, LSTAG, LEV, and sector (CONGS, INDGC and OILC) do not have any significant relationship with voluntary disclosure.

Wald test was conducted for the board attribute variables and the result is insignificant (p-value = 0.1346). The implication is that the board attribute variables do not help in explaining variation in voluntary disclosure in NLCs. This result contrasts the position of the agency theory and finds support for the information asymmetry theory.

4.3.2 Robustness tests

Several tests were conducted to examine the robustness of the findings. First, in the regression model, the sectors were excluded and the results remain qualitatively similar suggesting that the sector to which a firm belongs has no influence on the level of voluntary disclosure in NLCs. Second, the presence of block shareholders was used to replace their percentage shareholding in the equation. The result does not change except that the presence of block shareholders does not influence the level of voluntary disclosure.

Third, when ROE is used to substitute for ROA, CEO duality becomes significant at 10% level; performance lost its significance while other results remain unchanged. Fourth, the squared term of board composition was introduced into the model and the result does not show the existence of quadratic relationship between board composition and voluntary disclosure, while all other results remain similar.

Finally, following Krivogorsky (2006) multicollinearity test was conducted by computing the variance inflation factor (VIF) for all variables that have Pearson correlation coefficient greater than 0.30. The maximum VIF is 4.03 indicating that multicollinearity is not a problem in this study.

5 Conclusions

How do board attributes affect voluntary disclosure? From the agency theory perspective, a quality board should be able to do adequate monitoring of management actions to reduce information asymmetry between the shareholders and managers. This is because information asymmetry theory suggests that managers are not inclined to disclose information to shareholders. This study examines the effect of board attributes on the extent of voluntary disclosures using data for 237 firm-year observations from 75 firms listed on the NSE from 2009 to 2012. Findings show that board attributes do not have positive impact on the level of voluntary disclosure. In particular, firms with CEO duality show association with lower voluntary disclosure. This result suggests that CEOs that are chairmen of their firms compromise their boards with respect to the extent of information disclosure in the annual reports. Further, board size and board composition show insignificant relationship with voluntary disclosure. This supports the finding of La Porta

et al. (1998) that independent boards may not be able to influence disclosure policies in countries with weak investor protection rights. It is also consistent with the assertion by Hermalin and Weisbach (1998, p.96) that "Corporations are not governed by the process that corporate law would seem to imply". This result could be evidence of the unobservable complex processes that characterise board operations in Nigeria.

This study is limited to data extracted from non-financial firms. This could limit the generalisability of the results. Therefore, inclusion of additional years and firms could improve the findings. Furthermore, the study considered the board attribute variables without examining the motivations for actual board behaviour that may impact the results. The above limitations do not invalidate the findings of this study, but rather provides avenue to engage future researchers.

This study has several implications for future researchers, regulators and investors. First, further studies should examine whether other emerging economies will have similar results especially countries with weak institutions like Nigeria. Second, further studies should examine board behaviour to understand the complexities that take place within boards that affect their monitoring role. Third, the regulators need to enhance their enforcement and compliance mechanism to ensure the boards are not captives of the CEO. Four, the corporate governance code should make mandatory disclosure requirements in the annual reports on matters where there is no motivation by the management to make such disclosures considering the decrease in voluntary disclosure from 2010 to 2012. Finally, investors should acquaint themselves with the profile and antecedents of directors they nominate to their boards so as to have directors that will not become captives of the CEO.

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Notes

1 Grey directors are outside directors that are j-independent, affiliated or in business relationship with the firm (Conyon, 2014; Core et al., 1999).

Appendix 1

Voluntary disclosure checklist

A	*General information*	
	General information about economic environment	
	1	Information about the economy
	2	Information about the industry
	General corporate information	
	3	Brief History of firm
	4	Financial highlights statement
	5	Information about products
	6	Forward looking information
	7	Graphs and photos
B	*Non-financial information*	
	Information about directors and compensation committee matters	
	8	Age of directors
	9	Directors educational qualifications and experience
	10	Tenure of the CEO
	11	Information on existence of compensation
	12	Attendance at meetings of compensation committee
	13	Information on highest paid director
	Employee information	
	14	Information about equal employment opportunity
	15	Information about employees
	16	Information about staff training and development
	17	Data on accidents
	18	Safety measures
	Social policy	
	19	Information about corporate social responsibility
	20	Charitable donations
	21	Community programs
C	*Financial information*	
	Financial ratios	
	22	Performance ratios
	23	Profitability ratios
	24	Gearing ratios
	25	Liquidity ratios
	26	EPS

Voluntary disclosure checklist (continued)

	Financial review
27	Financial summary 5 or more years
28	Statement of accounting policies
29	Cash flows
	Market information
30	Share price at year end
31	Share price trend
32	Market capitalisation at year end
33	Market capitalisation trend
34	Size of share holdings
35	List of ten or more highest shareholders
36	Dividend policy